MongoDB Interview Questions & Answers

1. What is MongoDB and how is it different from SQL databases?

MongoDB is a NoSQL, document-oriented database that stores data in JSON-like documents (BSON). Unlike relational databases like MySQL, it has a flexible schema, no joins by default, and is built for scalability and performance. SQL databases use structured tables, fixed schema, and support joins. MongoDB is better for unstructured data and agile development.

2. Explain the aggregation pipeline with an example.

The aggregation pipeline processes data using stages like $match, $group, $sort, and $project. Example:

```
db.orders.aggregate([
  { $match: { status: 'delivered' } },
  { $group: { _id: '$customerId', totalAmount: { $sum: '$amount' } } },
  { $sort: { totalAmount: -1 } }
])
```

3. How do you design a schema for an e-commerce site in MongoDB?

Use embedded documents for frequently accessed data and references for large or related data.

Example:

Products: { name, price, category }

Users: { name, email, orders: [{ productId, quantity, date }] }

4. What is the difference between insertOne() and insertMany()?

insertOne() inserts a single document:

```
db.users.insertOne({ name: 'Aarav', age: 25 })
```

insertMany() inserts multiple documents:

```
db.users.insertMany([{ name: 'Diya' }, { name: 'Neha' }])
```

5. How does MongoDB handle transactions?

MongoDB supports ACID transactions from version 4.0+ using startSession(). It ensures that multiple operations succeed or fail together.

6. How do you scale a MongoDB database?

By using sharding. Sharding splits data across multiple machines using a shard key. This allows horizontal scaling and handling of large datasets.

7. What is the use of indexing, and how do you create one?

Indexes improve performance of read operations. Create an index using:

db.users.createIndex({ email: 1 })

8. What happens if the primary node in a replica set fails?

MongoDB performs automatic failover. A secondary node is elected as the new primary.

9. How does MongoDB achieve high availability?

Through replica sets, automatic failover, and redundancy. Data is copied across nodes, ensuring availability.

10. When would you choose MongoDB over MySQL?

Use MongoDB when you have unstructured data, need fast writes, flexible schema, or horizontal scalability. Use MySQL for structured data, complex joins, and strict consistency.